QUANTUM ERROR CORRECTION (QEC) — GRADUATE EXAM

Total Points: None | Questions: 5 | Date: March 28, 2026

Diagram-based questions on Quantum Error Correction (QEC)

Question 1

10.0 points

Analyze the circuit shown below of a stabilizer code. Identify how the syndrome measurement helps in error detection and explain the role of stabilizers in quantum error correction.

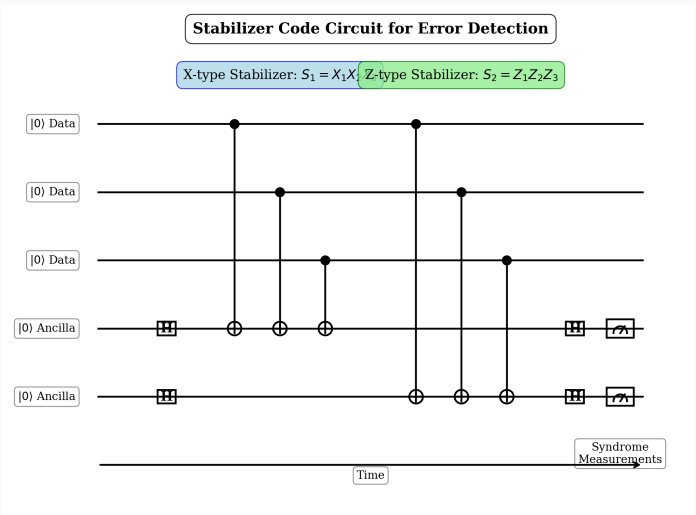

Fig. 1. Diagram for Question 1

Question 2

10.0 points

Examine the circuit shown below illustrating the 3-qubit bit-flip code with its corresponding error detection circuit. Explain the error correction process and identify the role of each circuit component in detecting a single bit-flip error.

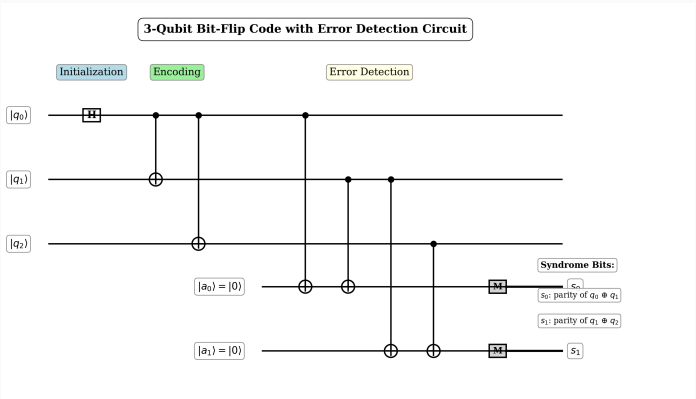

Fig. 2. Diagram for Question 2

Question 3

10.0 points

Review the surface code lattice diagram below. Describe how logical qubits are encoded on the lattice and discuss how the layout is used for error correction.

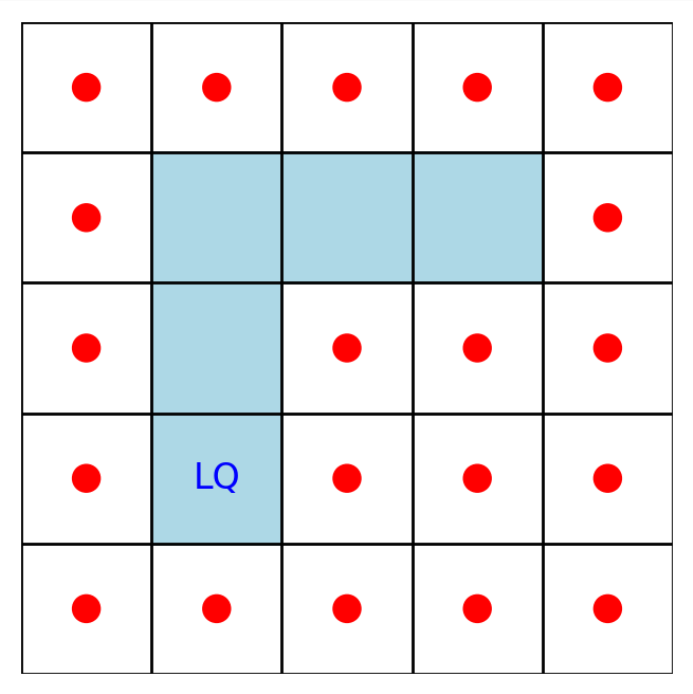

Fig. 3. Diagram for Question 3

Question 4

10.0 points

Analyze the diagram below demonstrating fault-tolerant gate teleportation. Explain how teleportation is used to achieve fault-tolerance in quantum gates and what protections it provides against errors.

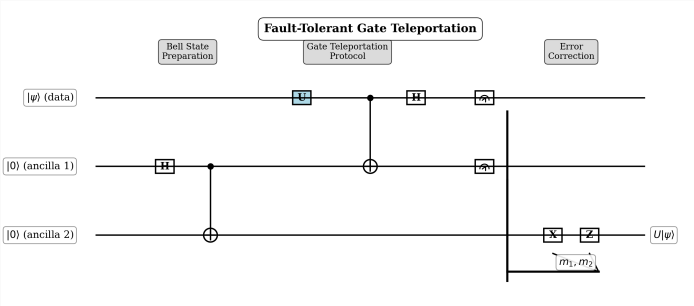

Fig. 4. Diagram for Question 4

Question 5

10.0 points

Interpret the threshold theorem in the diagram below, which illustrates error rates in quantum error correction. Explain how the error rate threshold is defined and its significance in achieving fault-tolerant quantum computation.

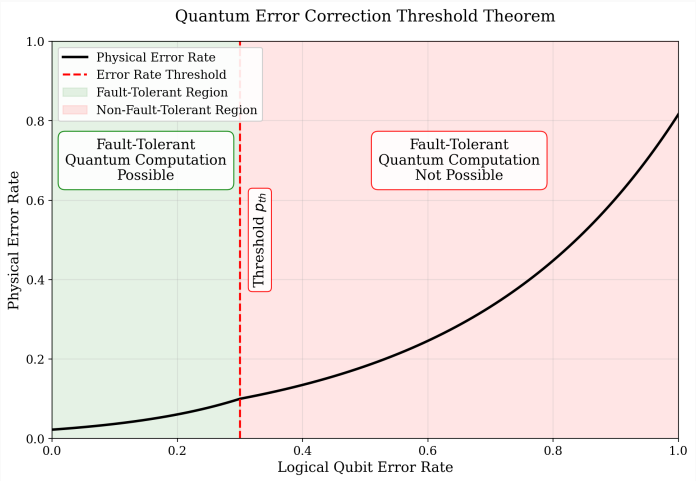

Fig. 5. Diagram for Question 5